SCHOOL OF DATA

Mini EDA Project

Housing Price Analysis

An Exploratory Data Analysis Report

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Dataset: Housing.csv — 545 residential property records

Introduction

Working with datasets has taught me that numbers rarely tell you the truth on their own, they tell you a story, but only if you are willing to sit with them long enough to ask the right questions. A dataset is not just a table of values; it is a compressed record of real decisions, real trade-offs, and real behaviour, and the job of analysis is to decompress that record carefully, without projecting assumptions onto it before the evidence has had a chance to speak. That is really what this project reinforced for me: how much of good analysis is simply discipline, checking the data before trusting it, and letting patterns emerge instead of hunting for the ones I expected to find.

With that mindset, I approached this housing dataset of 545 houses, each described by its price, area, number of bedrooms, bathrooms and stories, and a set of yes/no features, main road access, guest room, basement, hot water heating, air conditioning, and preferred-area status , along with a furnishing status of furnished, semi-furnished, or unfurnished. Going in, I expected price to be driven mostly by size and bedroom count, since that is the intuitive story most people tell themselves about real estate. Part of the value of this exercise was finding out how much of that intuition actually holds up once the data is examined properly.

Before jumping into analysis, I spent time simply inspecting the data, checking its shape, its data types, and whether anything looked mislabeled. I then moved into a cleaning phase: there were no missing values and no duplicate rows, which was a pleasant surprise. An IQR (Interquartile Range) check flagged 15 price outliers and 12 area outliers, but I chose to retain them, since they represent genuinely large or expensive houses rather than data-entry errors. I also confirmed that every categorical column contained only its expected values (for example, “yes” and “no” in the binary feature columns).

Key Findings

This is where my initial assumptions began to be tested against the evidence.

1. **Area and price are only moderately linked.** I expected a strong correlation, but it came out to 0.54, meaningful, but far from the whole picture of what determines price.

2. Air conditioning made a bigger difference than I anticipated. Houses with air conditioning average N6.01M compared to N4.19M without, a 43% gap arising from a single yes/no feature.

3. Location-type features carry real weight. Preferred-area houses average 33% more than others (N5.88M vs N4.43M), and mainroad houses average N4.99M compared to N3.40M for houses off the main road. This finding tracked most closely with what I already believed about real estate.

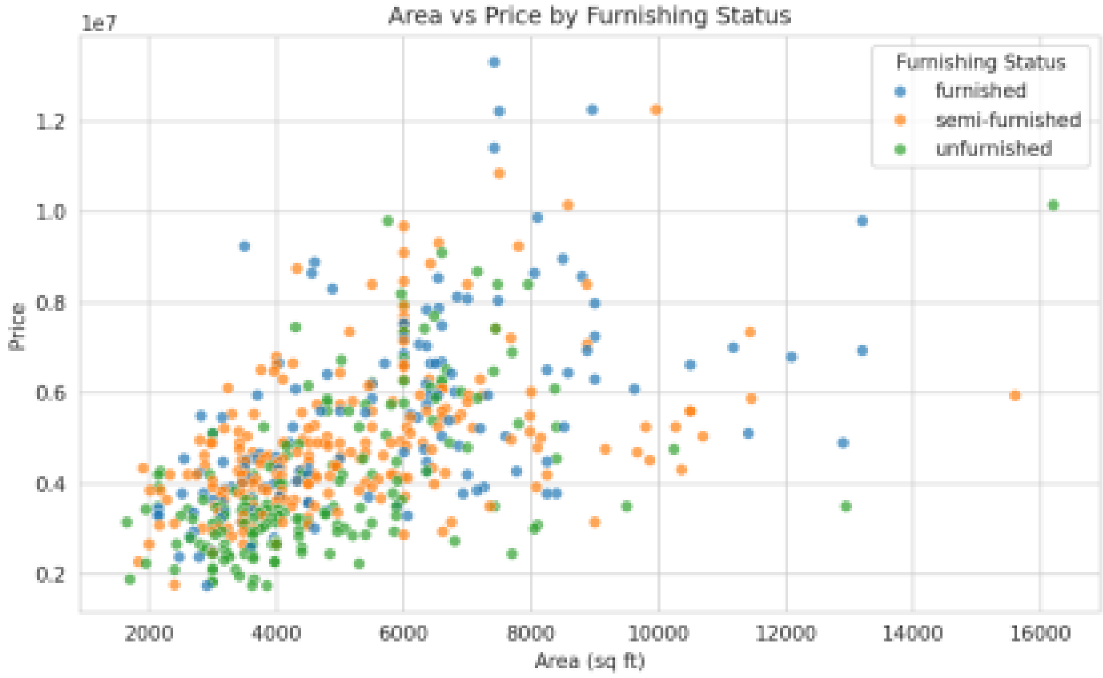
4. Stories and price climb together steadily. Average price rises from N4.17M (1 storey) to N4.76M (2 storeys), N5.69M (3 storeys), and N7.21M (4 storeys), a clean upward trend rather than noise.

5. Three-bedroom houses dominate the dataset, at 55% of all records. This makes it the effective "default" house size. Notably, houses with both a basement and air conditioning average N6.08M, higher than either feature alone would suggest, indicating that some features compound rather than simply add up.

The scatterplot of area against price, coloured by furnishing status, shows furnished houses clustering somewhat higher on the price axis. The correlation heatmap made it easy to see that price correlates most strongly with bathrooms (0.52), area (0.54), and air conditioning (0.45) , bedrooms, despite being the most "obvious" feature, correlate less strongly than expected.

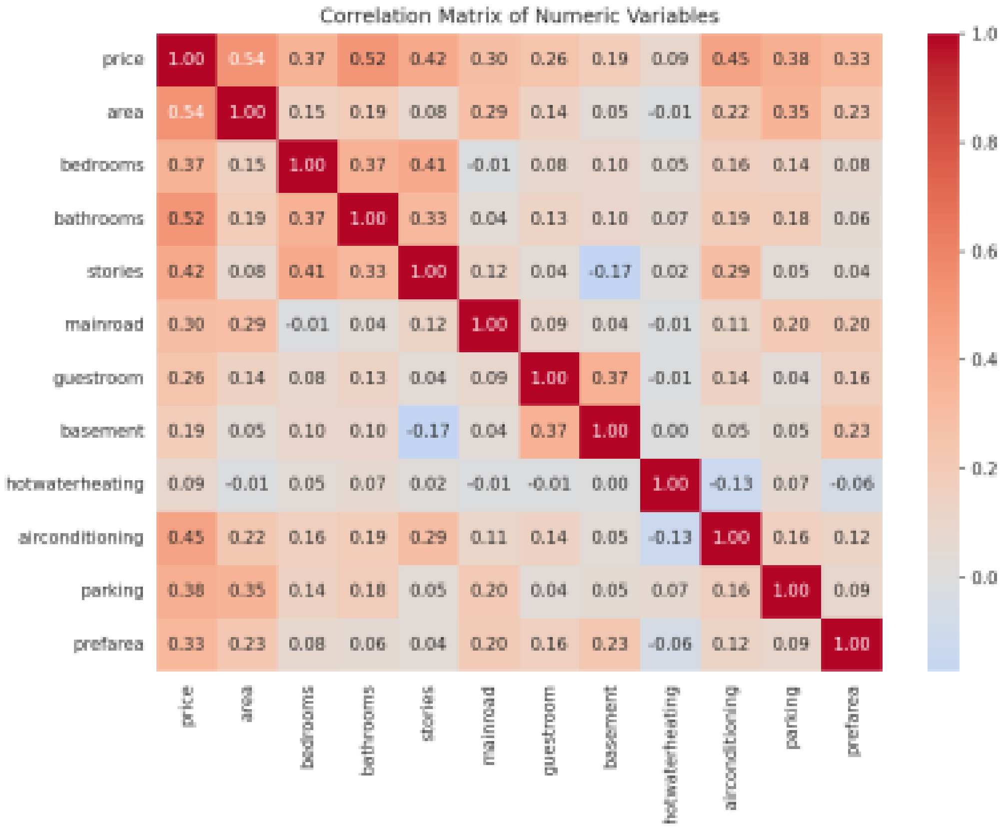
Supporting Visualisations

Figure 1: Area vs Price by Furnishing Status



This scatter plot shows the relationship between house area and price, coloured by furnishing status. Furnished houses tend to cluster toward the higher end of the price range at a given area, while unfurnished houses skew lower.

Figure 2: Correlation Matrix of Numeric Variables



This heatmap displays pairwise correlations between all numeric and encoded binary variables. Price
correlates most strongly with bathrooms, area, and air conditioning, while bedrooms show a comparatively weaker relationship with price.

Conclusion and Recommendations

If I were house-hunting with this data in hand, I would not fixate on bedroom count, as it moves the price far less than air conditioning, number of storeys, or location. Based on the findings above, I would offer the following recommendations to a hypothetical buyer looking for value:

• Prioritise location trade-offs consciously, houses off the main road or outside the preferred area are meaningfully cheaper (up to ~33–47% less) without necessarily sacrificing size or structure.

• Do not overpay purely for bedroom count, since 3-bedroom houses are already the market standard, extra bedrooms alone are not the strongest price driver and should not be the deciding factor.

• Consider furnishing the home independently rather than buying furnished, furnished houses average N5.50M versus N4.01M for unfurnished ones, a gap that likely exceeds the cost of furnishing it personally.

Overall, this project reinforced that price in this dataset is shaped by a combination of size, comfort features, and location, not any single variable in isolation. That, to me, is the real lesson of exploratory data analysis: resisting the urge to settle on the first plausible explanation, and instead letting the data reveal which of our assumptions actually hold.